

7

الكود \* (قد تمت مراجعة بيانات الطالب ورقم الجلوس)

DIVB

 **Will be reviewed**

8

The position of the vertex after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glTranslatef(-1.0f, -1.0f, 0.0f);  
glRotatef(90.0, 0, 0, 1.0);  
glBegin(GL_POINTS);  
glVertex3f(1.0, 0.0, 0.0);  
glEnd(); * 
```

☐ (0,1,0)

☒ (1,-1,0)

☐ (1,0,0)

☐ (-1,0,0)

 **Will be reviewed**

9

When is the timer callback function registered with glutTimerFunc invoked? \* 

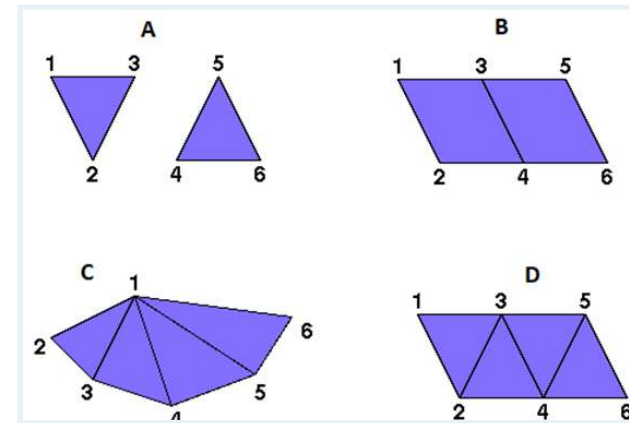
- ☐ After a specified interval, in milliseconds.
- ☐ When a key is pressed.
- ☒ Immediately after glutTimerFunc is called.
- ☐ When the window is resized.

 **Will be reviewed**

10

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_TRIANGLE_STRIP);  
glVertex2fv(v1); glVertex2fv(v2);  
glVertex2fv(v3); glVertex2fv(v4);  
glVertex2fv(v5); glVertex2fv(v6);  
glEnd();
```



☐ A

☐ B

☐ C

☒ D

**Will be reviewed**

11

What is the purpose of the glutDisplayFunc() callback? \*



☐ To handle keyboard input events.

- ☒ To handle window resizing events.
- ☐ To define the rendering code that is called whenever the window needs to be redrawn.
- ☐ To handle mouse input events.

 **Will be reviewed**

12

The OpenGL represents the following matrix as

\* 

$$m = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 1 & 2 & 3 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- ☐ `GLfloat m[] = { 1.0f, 2.0f, 3.0f, 4.0f, 5.0f, 6.0f, 7.0f, 8.0f, 1.0f, 2.0f, 3.0f, 4.0f, 0.0f, 0.0f, 0.0f, 1.0f };`
- ☐ `GLfloat m[] = { 1.0f, 5.0f, 1.0f, 2.0f, 6.0f, 2.0f, 3.0f, 7.0f, 3.0f, 4.0f, 8.0f, 4.0f};`
- ☒ `GLfloat m[] = { 1.0f, 5.0f, 1.0f, 0.0f, 2.0f, 6.0f, 2.0f, 0.0f, 3.0f, 7.0f, 3.0f, 0.0f, 4.0f, 8.0f, 4.0f, 1.0f };`
- ☐ `GLfloat m[] = { 1.0f, 2.0f, 3.0f, 4.0f, 5.0f, 6.0f, 7.0f, 8.0f, 1.0f, 2.0f, 3.0f, 4.0f};`

 **Will be reviewed**

13

What are the parameters typically passed to the glutReshapeFunc callback function? \* 

- ☐ The current time in milliseconds.
- ☐ The x and y coordinates of the mouse cursor.
- ☐ The state of the keyboard keys.
- ☒ The width and height of the window in pixels.

 **Will be reviewed**

14

What is the primary purpose of the glutReshapeFunc function in OpenGL? \* 

- ☒ To handle window resizing events.
- ☐ To handle
- ☐ To draw the scene.

☐ To initialize OpenGL.

 **Will be reviewed**

15

What is the primary purpose of culling in computer graphics? \* 

- ☐ To optimize the use of graphics hardware
- ☐ To reduce the frame rate of a graphics application
- ☐ To improve the visual quality of rendered images
- ☒ To reduce the number of polygons that need to be rendered

 **Will be reviewed**

16

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);  
glLoadIdentity();  
glRotatef(90.0, 0, 0, 1.0);
```

glScalef(-2.0, 1.0, -1.0);  
glRotatef(180.0, 0, 0, 1.0); \*

| A)                                                                                                 | B)                                                                                                  | C)                                                                                                 | D)                                                                                               |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| $\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ | $\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ | $\begin{bmatrix} 0 & 1 & 0 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ |

- ☐ A
- ☐ B
- ☐ C
- ☒ D

 **Will be reviewed**

17

What kind of argument is expected for glColor3fv command? \*

- ☒ An array typecasted with (GLvoid \*) or in the type indicated by the preceding letter.

- ☐ Any void variable
- ☐ A GLuint pointer
- ☐ A virtual method function pointer

 **Will be reviewed**

18

What do the parameters of glRectf represent? \* 

- ☒ The top-left and bottom-right corners of the rectangle.
- ☐ The width and height of the rectangle.
- ☐ The color of the rectangle.
- ☐ The center point and radius of the rectangle.

 **Will be reviewed**

19

The Modelview matrix after executing the following OpenGL code segment is:  
`glMatrixMode(GL_MODELVIEW);`

```
glLoadIdentity();
glRotatef(90.0, 0, 0, 1.0);
glScalef(-1.0, 2.0, -1.0); *
```

| A)                                                                                                 | B)                                                                                                  | C)                                                                                                 | D)                                                               |
|----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
| $\begin{bmatrix} 0 & -1 & 0 & 0 \\ -2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ | $\begin{bmatrix} 0 & -2 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ | $\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ | $\begin{bmatrix} 0 & 1 \\ 2 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$ |

- ☐ A
- ☒ B
- ☐ C
- ☐ D

 **Will be reviewed**

20

..... \* Rotate(90.0, 0, 0, 1.0)= Rotate(90.0, 0, 0, 1.0)\* Translate(2,-1,3) \*

- ☐ Translate(-2,1,3)

☐ Translate(2,-1,3)

☐ Translate(1,3,-2)

☒ Translate(1,2,3)

 **Will be reviewed**

21

What is the purpose of the glFlush() function in OpenGL? \* 

☐ To define the viewing volume

☐ To set the current color

☐ To clear the color buffer

☒ To force the execution of OpenGL commands

 **Will be reviewed**

22

Which of the following is NOT a valid mode for glPolygonMode? \* 

- ☐ GL\_FILL
- ☐ GL\_LINE
- ☐ GL\_POINT
- ☒ GL\_WIREFRAME

 **Will be reviewed**

23

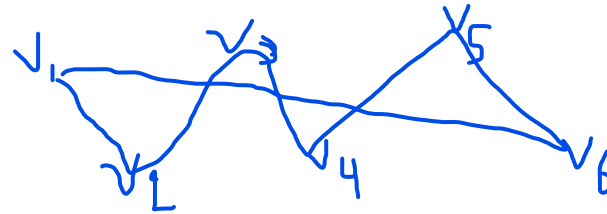
When should glutMainLoop() typically be called in an OpenGL program? \* 

- ☐ After drawing the scene.
- ☐ At the beginning of the program.
- ☒ After initializing OpenGL and setting up the display callback function.
- ☐ Before creating the window.

 **Will be reviewed**

Select the output of executing the following OpenGL code segment:

```
glBegin(GL_LINE_LOOP);  
glVertex2fv(v1); glVertex2fv(v2);  
glVertex2fv(v3); glVertex2fv(v4);  
glVertex2fv(v5); glVertex2fv(v6);  
glEnd(); * 
```



- ☐ A
- ☐ B
- ☒ C
- ☐ D
- ☐ E
- ☐ F

 **Will be reviewed**

What is the role of the glBegin and glEnd functions in relation to glVertex? \* 

- ☐ They set the projection and modelview matrices.

- ☐ They specify the color of the vertices.
- ☐ They control the depth testing and blending modes.
- ☒ They define the type of primitive to be drawn (e.g., points, lines, triangles).

 **Will be reviewed**

26

The Modelview matrix after executing the following OpenGL code segment is:

```
glMatrixMode(GL_MODELVIEW);
glLoadIdentity();
glRotatef(45.0, 0, 0, 1.0);
glRotatef(45.0, 0, 0, 1.0); *
```

**glRotatef(90.0, 0.0, 0.0, 1.0)**

- ☒ A
- ☐ B
- ☐ C
- ☐ D

$$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

 **Will be reviewed**

How many parameters does glViewport take? \*



- ☒ 4      `glViewport(GLint x, GLint y, GLsizei width, GLsizei height)`
- ☐ 2
- ☐ 5
- ☐ 3



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